
Deep Multimodal Fusion for Clinical Trial Outcome Prediction

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Abstract

Clinical trial failure leads to immense financial and scientific costs, yet accurately identifying at-risk trials remains a challenge due to the severe class imbalance in public registries. Standard predictive models frequently succumb to the “accuracy paradox,” maximizing global accuracy while failing to detect the minority failure class. Here, we propose a Multimodal Fusion Architecture utilizing an intermediate fusion strategy. We integrate structured metadata via a Multi-Layer Perceptron (MLP) with deep semantic text embeddings generated by BioClinical-ModernBERT. To address the critical imbalance (approx. 14% failure rate), we employ a random downsampling strategy on the training set. Our results demonstrate that this fusion approach recovers latent risk signals, achieving a Failure Recall of 0.68 and outperforming unimodal baselines. While this sensitivity incurs a trade-off in precision, the model functions effectively as a robust, high-recall screening tool for proactive risk assessment.

1 Introduction

Clinical trials represent the bottleneck of medical progress, consuming vast financial resources and years of operational effort [1]. While the aggregate failure rate of biomedical research is high, public trial registries are paradoxically dominated by completed studies, creating a severe class imbalance that obscures the signals of failure [2]. Traditional predictive approaches often rely on expert opinion or limited structured metadata (e.g., enrollment size), ignoring the rich semantic context hidden in trial protocols [3]. Furthermore, standard machine learning models trained on this imbalanced data frequently succumb to the “accuracy paradox,” maximizing global accuracy by ignoring the minority failure class entirely [4].

Scope and Problem Formulation. We utilize the `clinical-trials` dataset comprising approximately 540,000 records to predict definitive trial outcomes. We acknowledge that trial outcomes are often driven by external confounders—such as financial insolvency or regulatory shifts—that are absent from public metadata. Therefore, our objective is not to model causality, but to identify predictive risk signals embedded within the trial’s intrinsic scientific design (unstructured text) and operational parameters (structured metadata).

Contributions. To address the dual challenges of multimodal integration and class imbalance, this paper makes the following contributions:

- **Multimodal Architecture:** We propose an Intermediate Fusion neural network that integrates structured metadata with deep semantic embeddings generated by BioClinical-ModernBERT, capturing both operational and protocol-based risk factors.
- **Imbalance Mitigation:** We demonstrate that standard training fails to detect the minority class (Recall 0.00 - 0.44). We implement a strategic Random Downsampling strategy that recovers the minority signal, identifying risk factors that naive models miss.
- **Evaluation Framework:** We provide a rigorous comparative analysis, establishing that our fusion approach achieves a Failure Recall of 0.68, significantly outperforming unimodal baselines (Recall 0.00–0.44).

2 Related Work

Recent advancements in clinical trial outcome prediction have shifted from traditional statistical methods to deep learning approaches [5].

Semantic Representation. Early approaches treated trial protocols as simple "Bag-of-Words" or utilized standard topic modeling. While BioBERT [6] established the standard for biomedical transfer learning, it is constrained by standard token limits and older pre-training corpora. We adopt the recently released BioClinical-ModernBERT, leveraging its domain-specific optimization to extract richer, dense representations from the complex medical nomenclature found in eligibility criteria. Unlike SPOT [7], which models trials as sequential temporal events, our approach focuses on the static semantic risk encoded in the initial protocol design.

Multimodal Fusion. Effectively integrating high-dimensional text with low-dimensional meta-data remains a non-trivial challenge. Prior works like HINT [8] utilized hierarchical interaction graphs to model these relationships. While effective, graph networks introduce significant computational overhead. Sarigul et al. explored Attention-Based Fusion [9] to weigh modalities dynamically. In contrast, our work validates a streamlined Intermediate Fusion architecture. We demonstrate that projecting high-dimensional embeddings into a compact latent space combined with late-stage concatenation can achieve high sensitivity without the architectural complexity of GNNs or attention mechanisms.

Class Imbalance Strategies. The extreme class imbalance in trial registries is a persistent obstacle. While oversampling techniques like SMOTE [10] are common, we reject them for this domain because generating synthetic points between text embeddings can disrupt semantic integrity, creating vectors that do not correspond to valid trial protocols. Instead, our approach employs strict Random Downsampling of the majority class. We show that balancing the training distribution is sufficient to recover the minority signal, shifting the focus from global accuracy (which masks failure) to Failure Recall—a distinction often overlooked in general outcome prediction frameworks.

3 Methods

Dataset and Target Definition. We utilized the `louisbrulenaudet/clinical-trials` dataset, comprising approximately 540,000 trials. The predictive task is formulated as a binary classification problem. We define the positive class (Success) as trials with the status `COMPLETED`, and the negative class (Failure) as those marked `TERMINATED`, `WITHDRAWN`, or `SUSPENDED`. All other statuses (e.g., `RECRUITING`, `UNKNOWN`) were excluded. The resulting dataset exhibits severe class imbalance, with a distribution of approximately 86% Successes to 14% Failures.

Structured Feature Engineering. We selected seven structured features: *phases*, *study_type*, *enrollment_count*, *lead_sponsor_class*, *sex*, *minimum_age*, and *maximum_age*. Rows containing missing values for *sex* or *enrollment_count* (constituting < 2% of the dataset) were discarded. Remaining missing numerical values were imputed using the training set median, while missing categorical entries were treated as a distinct "Unknown" category. Numerical features were standardized using Z-score normalization to facilitate neural network training. During Exploratory Data Analysis (EDA), we identified extreme outliers in the age columns (e.g., max values > 6000). To prevent

these artifacts from distorting the feature scaling, we applied Winsorization (clipping), capping age values at 120 years. Categorical features were processed via one-hot encoding.

Splitting Strategy. To prevent data leakage and ensure a realistic evaluation, we employed a stratified 70/15/15 split (Train/Validation/Test). Crucially, this partitioning was performed *prior* to any data balancing. The Validation and Test sets were maintained in their original, imbalanced state to accurately reflect the real-world prevalence of trial failure (14%) during evaluation.

Training Data Balancing. To mitigate the bias toward the majority class, we implemented a random downsampling strategy applied exclusively to the *Training* set. We selected downsampling over upsampling to avoid the severe overfitting associated with repeating minority text samples and to preserve the semantic integrity of the embedding space, which can be degraded by synthetic interpolation methods such as SMOTE. Additionally, we prioritized downsampling over class-weighted loss functions to significantly reduce the memory overhead of the 1536-dimensional embeddings, ensuring the deep learning pipeline remained computationally tractable on the available hardware. We adopted a comparative approach: initial baselines were trained on the raw, unbalanced data to quantify the baseline bias (Phase 1). Subsequently, we generated a balanced (50/50) training distribution by randomly downsampling the majority Success class to match the minority Failure class. This balanced set was utilized for the improved Random Forest baselines (Phase 2) and the final Multimodal Fusion model.

Feature Reconstruction and Model Selection. The original dataset documentation indicated the presence of pre-computed embeddings; however, these features were absent from the distributed files. To reconstruct the missing embeddings, we implemented a custom generation pipeline using BioClinical-ModernBERT-base, as recommended by the dataset creators. This Transformer-based model is pre-trained on large-scale biomedical corpora, allowing it to capture domain-specific semantic nuances—such as complex medical terminology in eligibility criteria—that generic models (e.g., BERT-base) often fail to represent.

Extraction Pipeline. Processing textual data for approximately 540,000 trials presented significant computational challenges. To prevent memory overflows and manage runtime constraints, we implemented a sharded inference strategy. The dataset was partitioned into chunks and processed in parallel using GPU acceleration (NVIDIA L4). For each trial, we encoded the `brief_summary` and `eligibility_criteria` fields into dense 768-dimensional vectors. These were subsequently concatenated to form a single 1536-dimensional semantic feature vector, effectively restoring the deep text modality required for the fusion architecture.

Models and Hyperparameters. We evaluate Random Forest classifiers ($n_{\text{estimators}} = 100$) to establish non-linear baselines for both data modalities. To prevent overfitting, particularly in the high-dimensional embedding space, we imposed maximum depth constraints ($d = 15$ for structured data, $d = 10$ for text embeddings).

Experimental Setup. We use a comparative two-phase strategy to understand the impact of data balancing. **Phase 1** models are trained on the raw, unbalanced data to show the baseline bias. **Phase 2** models are trained on the downsampled (balanced) dataset to quantify the improvement in minority class detection.

Evaluation Metrics. Given the risk asymmetry in clinical trials, where a false negative (missed failure) causes significantly higher economic and time costs compared to a false positive, we chose Failure Recall (Sensitivity of the minority class) as our primary performance metric. We deliberately avoid global metrics such as Macro or Weighted F1-scores, as these are inflated by the majority class. We report class-specific Precision and F1-Scores for the minority class to characterize the trade-off in false alarms. Global accuracy is reported for reference but is treated as secondary due to the inflation caused by the majority class (the “Accuracy Paradox”).

Intermediate Fusion Strategy. To integrate the different data modalities (low-dimensional structured metadata (approx. 23 features) and high-dimensional semantic embeddings (1536 features)) we propose an Intermediate Fusion neural network. Unlike early fusion (concatenation at the input level), which would allow the high-dimensional embeddings to mathematically overwhelm the

structured signal, our architecture processes each modality through independent sub-networks to extract dense latent representations before integration.

Branch Topologies. The architecture consists of two parallel encoders:

- **Structured Branch:** A shallow multi-layer perceptron (MLP) projects the standardized metadata into a 64-dimensional latent space. We utilize Batch Normalization and ReLU activation to capture non-linear interactions between trial design parameters (e.g., *Phase* vs. *Sponsor*).
- **Text Branch:** To handle the high-dimensional input (1536d), we employ a **bottleneck architecture** that progressively compresses the semantic signal (1536→512→128). Due to the propensity of high-dimensional embeddings to overfit on small datasets, we apply aggressive regularization (Dropout $p = 0.5$) within this branch.

Fusion and Inference. The outputs of the two branches are concatenated to form a single 192-dimensional multimodal vector. This combined representation is passed through a final fully connected classification head (192→64→1) to generate a single scalar logit representing the probability of trial success.

Training Protocol. The model was trained using Binary Cross Entropy with Logits Loss (BCEWithLogitsLoss) to ensure numerical stability. We utilized the Adam optimizer (learning rate 1×10^{-3}) with a batch size of 128. Training was conducted for 10 epochs on an NVIDIA L4 GPU. We monitored validation loss on the unbalanced validation set to detect overfitting, given the synthetic nature of the balanced training distribution.

4 Results

Comparison and results between model performances can be found in table 1

Table 1: Comparative performance on the held-out unbalanced Test Set. All non Accuracy metrics are based on Failure Class. Failure Recall is the primary metric of interest. Note the progression from high-accuracy/zero-recall in Phase 1 to high-sensitivity in the Fusion model. All results reported at a standard decision threshold of 0.50.

Model	Recall (Fail)	Precision (Fail)	F1 (Fail)	Accuracy
<i>Phase 1: Unbalanced Training</i>				
Structured Baseline (Model 1)	0.44	0.87	0.58	0.91
Embeddings Baseline (Model 2)	0.00	0.00	0.00	0.86
<i>Phase 2: Balanced Training</i>				
Structured Baseline (Model 3)	0.66	0.46	0.54	0.84
Embeddings Baseline (Model 4)	0.65	0.24	0.35	0.66
<i>Phase 3: Multimodal</i>				
Multimodal Fusion	0.68	0.25	0.37	0.67

Imbalanced Dataset (Phase 1). Focusing on our primary metric of Failure Recall, the initial baselines trained on raw data demonstrate the severity of the class imbalance (Appendix Figure 2, top row). The Unbalanced Embeddings model (Model 2) represents the extreme case of the “Accuracy Paradox”: despite achieving a global accuracy of 86%, it failed to identify a single failed trial (**Recall 0.00**). Similarly, the Structured Baseline (Model 1) achieved 91% accuracy but missed 56% of all actual failures (Recall 0.44). These results confirm that without intervention, standard classifiers default to a majority-class prediction strategy, effectively ignoring the minority risk cases.

Balancing Strategy (Phase 2) The application of random downsampling to the training set successfully recovered the predictive signal for the minority class (Appendix Figure 2, bottom row). The Structured Baseline (Model 3) improved its Recall from 0.44 to 0.66, but the most dramatic shift occurred in the text embeddings (Model 4), where Failure Recall surged from 0.00 to 0.65.

This proves that the trial descriptions contain significant latent risk indicators that were previously suppressed. However, this sensitivity came at a calculated cost: the Balanced Embeddings model (Model 4) dropped to a Precision of 0.24. This trade-off—accepting a higher rate of false alarms to minimize missed failures—is strategically aligned with our goal of risk detection.

Multimodal Synergy (Phase 3). The Multimodal Fusion model (Model 5) achieved the highest sensitivity of all experiments, with a Failure Recall of 0.68. By integrating the structured metadata with the semantic text representations via the intermediate fusion architecture, the model identified risk factors that unimodal baselines missed. It outperformed the best structured model by 2 percentage points and the text model by 3 percentage points. While the gain is small, it shows that the modalities contain complementary rather than redundant information; the text branch captures semantic nuance (e.g., inclusion criteria strictness) that the structured branch cannot access, and vice versa.

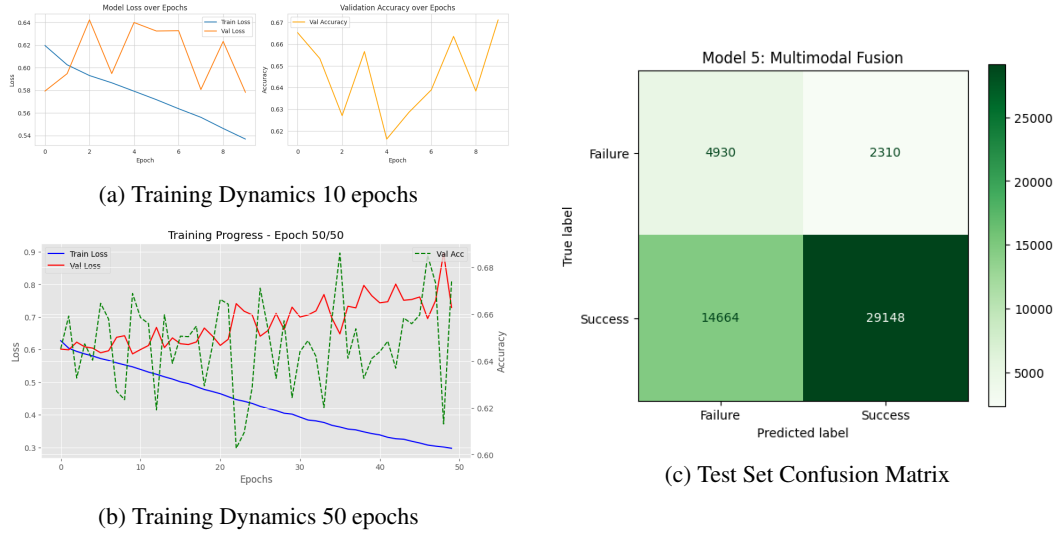


Figure 1: Performance analysis of the Multimodal Fusion Model. **(a)** Training loss (blue) converges smoothly, indicating stable feature learning, while validation loss (orange) remains more volatile due to the distributional mismatch between the balanced training set and the unbalanced validation set. **(b)** Model output after 50 epochs of training (blue = training loss, red = validation loss, green = validation accuracy). Training loss and validation loss diverges early (epoch 8), indicating overfitting from there on and out. **(c)** The model prioritizes sensitivity, correctly identifying 68% of failures (4,930 trials) at the cost of an elevated false-positive rate, supporting its use as a conservative screening tool.

Visual inspection of the training dynamics (Figure 1a) reveals a steady decrease in training loss, indicating successful feature learning. The validation metrics exhibit volatility due to the distributional shift between the balanced training set and the unbalanced validation set; however, the metrics do not diverge, validating our decision to stop training at 10 epochs to prevent overfitting. Training for 50 epochs confirmed that overfitting happened around the 10 epoch mark. Figure 1b details the classification behavior: the model prioritizes sensitivity, correctly flagging 68% of failures, but incurs a high false-positive rate (Precision 0.25). This aligns with our design goal, confirming the model’s utility as a conservative “early warning” screening tool rather than an autonomous decision-maker.

Post-Hoc Analysis (Thresholds) We investigated whether adjusting the decision threshold could improve the low Precision (0.25) of the Fusion model without sacrificing sensitivity. Since the model outputs the probability of success, lowering the acceptance threshold (e.g., classifying as “Success” even if probability is only > 0.30) reduces False Positives. Our analysis confirms that while this strategy improves Precision (rising to 0.34 at threshold 0.30), it causes a catastrophic collapse in Failure Recall (dropping to 0.34). Consequently, we retained the default threshold of 0.50, as maximizing the detection of failed trials (Recall 0.68) takes precedence over minimizing false alarms in a risk-screening context.

5 Discussion

Interpretation of Multimodal Signal The performance advantage of the Fusion architecture supports the hypothesis that structured metadata and text embeddings provide complementary information signals. We hypothesize that the structured features (e.g., *Sponsor*, *Phase*) describe the operational framework of the trial, capturing macro-level indicators of funding status and study scale. In contrast, the text embeddings derived from the `brief_summary` and `eligibility_criteria` encode the scientific logic of the design. The Fusion model likely succeeds by resolving conflicting signals between these modalities. For instance, a trial may appear operationally robust (e.g., an Industry-sponsored Phase 3 study) yet fail due to overly restrictive inclusion rules hidden in the unstructured text. By integrating these channels, the model identifies specific failure modes that neither modality could detect in isolation.

Operational Viability (The Precision Defense) We acknowledge that a Precision of 0.25 implies that the model generates approximately three false alarms for every correctly identified failure. While this rules out autonomous decision-making, it remains operationally viable given the medical domain’s extreme cost asymmetry. The sunk cost of a failed clinical trial dwarfs the administrative cost of a human expert reviewing a “false positive” protocol [1]. Consequently, we propose deploying the Fusion model as a human-in-the-loop triage system. By flagging high-risk protocols, the model acts as a preliminary filter, ensuring that expensive human oversight is allocated to the specific subset of trials where the probability of failure is highest.

Limitations, Confounders and Future Works A primary limitation of this study is the restriction to intrinsic trial data (protocol and metadata). Clinical trial failure is often driven by external factors invisible to the model, such as sudden regulatory shifts, loss of funding, or strategic corporate mergers. Consequently, our model captures predictive correlations within the study design rather than isolating causal drivers of failure. Future iterations could integrate financial time-series to control for these confounders. Methodologically, our use of random downsampling to address class imbalance discarded the majority of ‘Success’ samples, likely contributing to early overfitting due to data scarcity. Future work should instead train on the full dataset using **Weighted Loss**. By assigning a disproportionate penalty (e.g., 10x) to missed failures, the model could learn from the full data distribution while still prioritizing sensitivity.

While ModernBERT embeddings capture semantic similarity, they suffer from the “bottleneck” of compressing complex inclusion/exclusion logic into a fixed-size vector. Future work should transition from discriminative representation learning to generative reasoning using Large Language Models (LLMs). **Context Window:** Clinical protocols often exceed the token limits of standard BERT models, leading to truncation of critical details. Modern LLMs with extended context windows could ingest the full protocol. **Logical Inference:** Eligibility criteria often contain complex Boolean logic (e.g., “Criteria A AND B, UNLESS C”). While embeddings struggle to represent these logical dependencies, an LLM could be fine-tuned via Chain-of-Thought (CoT) prompting to explicitly detect logical contradictions or overly restrictive clauses that limit recruitment, potentially improving Precision.

6 Conclusion

This project demonstrates that the primary bottleneck in predicting clinical trial failure is not a lack of signal, but the suppression of that signal by severe class imbalance. We showed that standard unimodal classifiers fall to the “accuracy paradox,” achieving high accuracy but zero sensitivity to risk. By implementing a downsampling data balancing strategy and a Multimodal Fusion architecture, we successfully recovered these latent signals, increasing Failure Recall from naive baselines of 0.00–0.44 to a final 0.68. While this sensitivity suffers a trade-off in precision, the resulting model functions effectively as a high-recall screening tool. We conclude that integrating structured operational data with semantic protocol analysis offers a viable path toward proactive risk management, capable of flagging nearly 70% of doomed trials before they incur significant financial and scientific costs.

7 References

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8 Appendix A

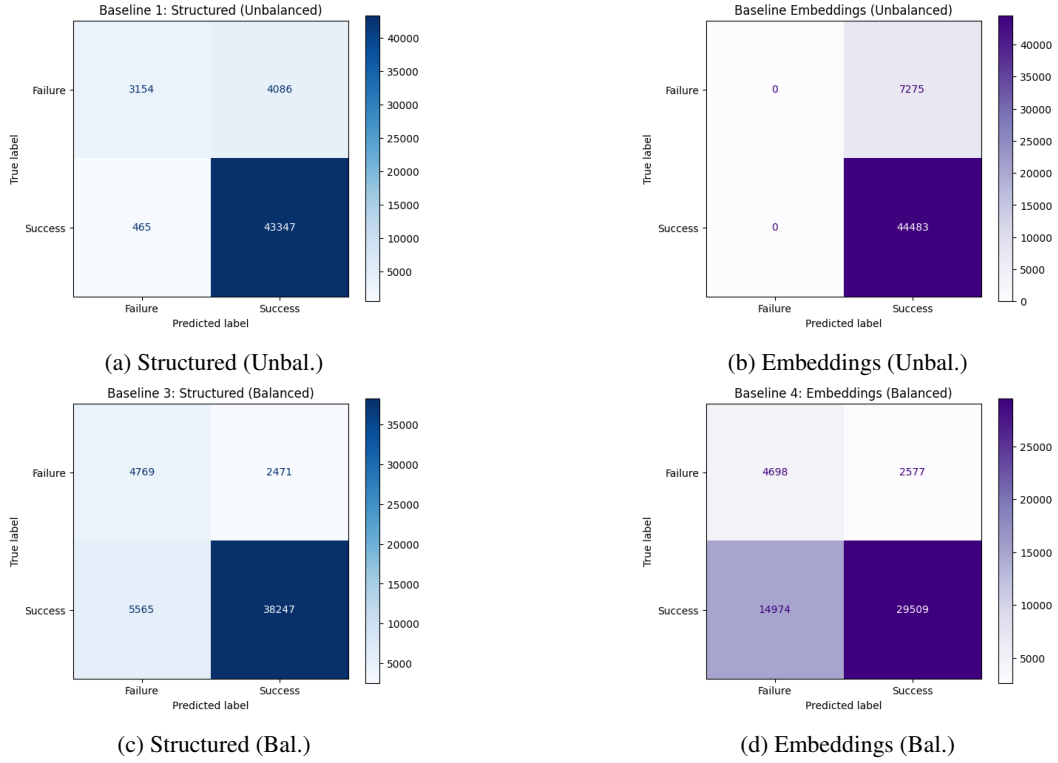


Figure 2: Confusion Matrices comparing Unbalanced (Top) vs. Balanced (Bottom) training strategies. Top Row: Models default to the majority class, resulting in near-zero Recall for failures (top-left quadrant). Bottom Row: Downsampling the training data forces the models to recognize the minority class, significantly increasing True Negatives (bottom-left) at the cost of increased False Positives (bottom-right)